
Envy-Free Chore Allocation with Unit Subsidies

under Negative Dichotomous Valuations

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Outline

- ① The problem
- ② Model
- ③ The construction
- ④ Two limitations
- ⑤ Summary

Envy-freeness, and why it fails

We allocate a set M of m indivisible items among n agents.

Envy-freeness [Foley 1967; Varian 1974]

An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is *envy-free* if

$$v_i(A_i) \geq v_i(A_j) \quad \text{for all } i, j.$$

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For *indivisible* items this is hopeless in general:

two agents, one item that both want.

Whoever does not receive it envies the one who does.

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Two standard escapes:

- **Relax** the criterion — EF1, EFX [Budish 2011; Lipton et al. 2004; Caragiannis et al. 2019]
- **Keep** exact envy-freeness, and pay for it. ← this talk

Envy-freeness with subsidies

Each agent i additionally receives a subsidy $p_i \geq 0$ from a third party.

Envy-free solution

$(\mathcal{A}, p)$ is an *envy-free solution* if

$$v_i(A_i) + p_i \geq v_i(A_j) + p_j \quad \text{for all } i, j.$$

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The question is not existence — it is **how much money**.

Throughout, each item is worth at most one unit to any agent; some normalisation of this kind is needed before absolute subsidy bounds mean anything.

What is known — for goods

- **Halpern and Shah (2019).** Every instance admits an envy-freeable allocation. For a *given* such allocation, total subsidy $(n - 1)m$ may be needed. They conjecture $n - 1$ suffices when the allocation may also be chosen.

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The last bound is tight: n agents, one good wanted by all — the $n - 1$ agents who miss out each need one unit.

This talk: chores

Now every item is a *chore* — receiving one weakly *decreases* utility.

Negative dichotomous valuations

For every agent i , every $S \subseteq M$ and every $e \notin S$,

$$v_i(S \cup \{e\}) - v_i(S) \in \{-1, 0\}.$$

The sign analogue of the dichotomous goods model. **No further structure is assumed** — valuations need not be additive, submodular, or even subadditive.

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Why binary marginals are natural here. An extra task either creates a fresh unit of burden, or is absorbed by a commitment already made:

- an agent already covering an on-call window absorbs another incident in that window at no extra cost;
- a driver already routed through a district absorbs one more stop.

Whether a chore is costly depends on the rest of the bundle — hence non-additivity.

Main result

Theorem

For every instance with negative dichotomous valuations there exist a *complete* allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with

$$\sum_{i \in N} p_i \leq n - 1$$

such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover $\mathcal{A}$ is *EF1 even before subsidies are paid*, and both are computable in polynomial time in the value-oracle model.

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- the per-agent bound is **0 or 1** — not merely $O(1)$;
- the total bound $n - 1$ is **tight**;
- EF1 is **extra** — the goods-side construction does not give it.

Where the result sits

	goods	chores
additive	Brustle et al. 2020	Lu et al. 2026
dichotomous	Barman et al. 2022	this work

Each cell gives one unit per agent and $n - 1$ in total.

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The two classes are **incomparable**: dichotomous valuations need not be additive, and additive valuations need not be dichotomous. So neither neighbouring cell implies ours.

The benchmark we had to beat

The best previously known bound for our domain is not from the chores literature at all, but from a much more general theorem.

Kawase, Makino, Sumita, Tamura, Yokoo (2025)

For *doubly monotone* valuations with marginals bounded by one in absolute value, any EF1 allocation can be converted into an envy-free solution with

$$n - 1 \text{ per agent,} \quad n(n - 1)/2 \text{ in total.}$$

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We improve $n - 1$ per agent to **1** per agent — a linear-factor improvement — and $n(n - 1)/2$ to $n - 1$ in total.

The total bound $n - 1$ is tight

Take n agents, $n - 1$ chores, and identical *binary additive* costs:

$$v_i(S) = -|S| \quad \text{for every agent } i.$$

In any complete allocation some agent h receives the empty bundle. Envy-freeness of agent i towards h reads

$$-|A_i| + p_i \geq v_i(A_h) + p_h = p_h,$$

hence

$$p_i \geq |A_i| + p_h \geq |A_i|.$$

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$$p_i \geq |A_i| + p_h \geq |A_i|.$$

Summing over all agents,

$$\sum_{i \in N} p_i \geq \sum_{i \in N} |A_i| = n - 1.$$

So no construction can do better in total.

□

Model and cost form

M : m indivisible chores. $N = [n]$: agents. $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$.

Algorithms run in the *value-oracle* model: an oracle returns $v_i(S)$ for a queried S .

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It is convenient to pass to non-negative *costs*

$$c_i(S) := -v_i(S).$$

Dichotomous cost function

$c(\emptyset) = 0$ and $c(S \cup \{e\}) - c(S) \in \{0, 1\}$ for all S and $e \notin S$.

Negative dichotomous valuations $\leftrightarrow$ dichotomous costs, under $c_i = -v_i$. In particular every c_i is **non-decreasing** and **integer-valued** — both used constantly below.

Write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal.

Fairness notions, in cost form

Envy-freeness

$c_i(A_i) \leq c_i(A_j)$ for all i, j .

Envy-free solution

$c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j .

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EF1 for chores

A complete $\mathcal{A}$ is *EF1* if for all i, j either

$$c_i(A_i) \leq c_i(A_j), \quad \text{or} \quad \exists e \in A_i : c_i(A_i \setminus \{e\}) \leq c_i(A_j).$$

The item removed comes from the *envious agent's own* bundle — the natural form for chores, and equivalent to the usual two-sided definition on all-chore instances.

A partial allocation X has *residue* $R(X) := M \setminus \bigcup_i X_i$.

The equality graph

Given a partial allocation X , define the *equality graph* $G = (N, E)$ on the agents:

$$(i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j).$$

An arc $i \rightarrow j$ records that agent i is *exactly indifferent* between her own bundle and j 's — she is on the verge of envying j .

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Strongly connected component; tail component

u, v are *strongly connected* if $u = v$, or each is reachable from the other. The equivalence classes are the *strongly connected components* (SCCs).

An SCC S is a *tail component* if no arc leaves it: there is no $(i, j) \in E$ with $i \in S$ and $j \notin S$.

Tail components exist

Lemma

Every finite non-empty directed graph has at least one tail strongly connected component.

PROOF. Start at any SCC. If it is not a tail component, follow an outgoing arc to another SCC and repeat.

No SCC can be visited twice: that would make the two components mutually reachable, hence a single component.

There are finitely many SCCs, so the walk terminates — at a component with no outgoing arc, i.e. a tail component. □

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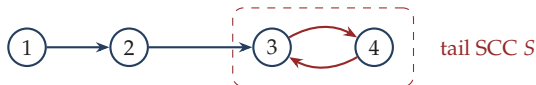
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Starting point: an envy-free *partial* allocation

Theorem [Tao, Wu, Yu, Zhou 2025, Thm. 6.1]

For every n and all dichotomous costs $c_1, \dots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free,

$$c_i(X_i) \leq c_i(X_j) \quad \text{for all } i, j,$$

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and leaves $|R(X)| \leq n - 1$ chores unassigned.

We need more than the statement — we need the **state the algorithm halts in**. So we recall its three rules.

It maintains X together with the equality graph G , and while $R(X) \neq \emptyset$ applies, *in order*:

The three rules

- (R1) If some $e \in R(X)$ and some i have $c_i(e \mid X_i) = 0$ — a *free* chore for i — assign e to i .
- (R2) Otherwise, if some $e \in R(X)$ and some arc $(i, j) \in E$ on a *directed cycle* of G have $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i .
- (R3) Otherwise, select a tail SCC S of G . If $|R(X)| \geq |S|$, hand out distinct residual chores to the agents of S , one each. Otherwise **halt**.

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- (R3) Otherwise, select a tail SCC S of G . If $|R(X)| \geq |S|$, hand out distinct residual chores to the agents of S , one each. Otherwise **halt**.

So if the algorithm halts with residue left over:

- neither (R1) nor (R2) applies at that state, and
- for the tail SCC S selected at the halting step, $r := |R(X)| < |S|$.

That last inequality is what gives us *room*: more agents in S than chores left to place.

What the terminal state gives us

Fix X at termination; $R := R(X)$, $r := |R|$, G the equality graph there, and S the tail SCC from the halting step (so $r < |S|$).

Lemma (terminal state)

Suppose $r \geq 1$. Then

- (i) $c_i(e \mid X_i) = 1$ for every $i \in N$, $e \in R$;
- (ii) $c_i(e \mid X_j) = 1$ for every arc $(i, j) \in E$ with $i, j \in S$, and every $e \in R$;
- (iii) $c_i(X_i) < c_i(X_j)$ for every $i \in S$, $j \in N \setminus S$.

In words:

- (i) no leftover chore is free to anybody;
- (ii) inside S , no leftover chore is free even on a neighbour's bundle;
- (iii) agents in S are *strictly* better off than everyone outside S .

Proof of the terminal lemma

Since $r \geq 1$, the algorithm halted with non-empty residue. Hence neither (R1) nor (R2) was applicable.

(i) If $c_i(e \mid X_i) = 0$ for some i and $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$, so $c_i(e \mid X_i) = 1$.

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- (ii) Let $(i, j) \in E$ with $i, j \in S$. As S is strongly connected there is a directed path $j \rightsquigarrow i$; take one without repeated vertices and append the arc (i, j) — a directed cycle through (i, j) .

If $c_i(e \mid X_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and that cycle. So the marginal is not 0, hence 1.

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(iii) Let $i \in S, j \notin S$. As S is a *tail* component no arc leaves it, so $(i, j) \notin E$, i.e.

$$c_i(X_i) \neq c_i(X_j).$$

Envy-freeness of X gives $c_i(X_i) \leq c_i(X_j)$. The inequality is therefore strict. □

The completion

Assume $r \geq 1$ and write $R = \{e_1, \dots, e_r\}$.

Since $r < |S|$, we may pick r *distinct* agents

$$T = \{t_1, \dots, t_r\} \subseteq S,$$

arbitrarily, and define

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \quad (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases}$$

The t_k are distinct and each chore of R is used once, so the bundles stay disjoint and now cover M : **$\mathcal{A}$ is complete.**

Agents in T are the *recipients* — each absorbs exactly one leftover chore, and by (i) that costs her exactly +1.

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What remains: decide *who gets paid*, and show one unit each is enough.

Who gets paid: the subsidy set

Definition (subsidy set)

Let $P \subseteq N$ be the *smallest* set with

(P1) $T \subseteq P$; and

(P2) if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.

Then set $p_i := 1$ if $i \in P$, and $p_i := 0$ otherwise.

Equivalently: start from $P := T$ and repeatedly absorb any agent *outside* S with an equality arc *into* P — a **backward closure** in G .

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Equivalently: start from $P := T$ and repeatedly absorb any agent *outside* S with an equality arc *into* P — a **backward closure** in G .

The intuition. A recipient's cost rose by one, so she needs a unit. Paying her may make her *neighbours* envious — exactly those who were indifferent to her, i.e. those with an equality arc into her. The payment must therefore propagate backwards.

The restriction $i \notin S$ in (P2) stops the closure from swallowing S itself — and that is what keeps the budget below n .

The algorithm

Step 2 is due to Tao et al.; steps 5–10 are the contribution

Input: agents $N = [n]$, chores M , value oracles $\{v_i\}$.

Output: complete $\mathcal{A}$ that is EF1, and $p \in \{0, 1\}^n$ with $(\mathcal{A}, p)$ envy-free, $\sum_i p_i \leq n - 1$.

1. Set $c_i = -v_i$ for every i .
2. Run the partial-allocation algorithm of Tao et al. on $\{c_i\}$, keeping the tail SCC selected at the terminating application of (R3) if the residue is non-empty.
3. Let X be its terminal partial allocation, $R = M \setminus \bigcup_i X_i$.
4. **if** $R = \emptyset$ **then return** $(X, 0)$.
5. Build the terminal equality graph $G = (N, E)$.
6. Write $R = \{e_1, \dots, e_r\}$; let S be that tail SCC. $r < |S|$
7. Choose distinct $T = \{t_1, \dots, t_r\} \subseteq S$.
8. $A_{t_k} \leftarrow X_{t_k} \cup \{e_k\}$ for $k \leq r$; $A_i \leftarrow X_i$ for $i \notin T$.
9. $P \leftarrow T$; **while** $\exists i \in N \setminus S, j \in P$ with $(i, j) \in E, i \notin P$: $P \leftarrow P \cup \{i\}$.
10. $p_i \leftarrow \mathbf{1}[i \in P]$; **return** $(\mathcal{A}, p)$.

The closure does not eat into S

Lemma

$P \cap S = T$. Equivalently $P \setminus T \subseteq N \setminus S$: every subsidised agent inside S is a recipient.

PROOF. $T \subseteq P$ by (P1) and $T \subseteq S$, so $T \subseteq P \cap S$.

Suppose for contradiction some $i \in (P \cap S) \setminus T$, and put $P' := P \setminus \{i\}$.

P' still satisfies (P1), since $i \notin T$.

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P' also satisfies (P2). Let $k \in N \setminus S$ with $(k, j) \in E$ for some $j \in P'$. Since $P' \subseteq P$ and P satisfies (P2), $k \in P$. Moreover $k \neq i$, because $k \notin S$ whereas $i \in S$. Hence $k \in P'$.

So P' satisfies both conditions and is strictly smaller than P — contradicting minimality. □

The budget

$$p \in \{0, 1\}^n \text{ and } \sum_{i \in N} p_i \leq n - 1.$$

PROOF. $p \in \{0, 1\}^n$ is immediate from the definition.

We have $|T| = r < |S|$, so $S \setminus T \neq \emptyset$.

By the previous lemma $P \cap S = T$, so every agent of $S \setminus T$ lies *outside* P and therefore receives **zero** subsidy.

At least one agent is unpaid, hence

$$\sum_{i \in N} p_i = |P| \leq n - 1.$$

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The slack came from $r < |S|$ — the halting condition of (R3).

The key lemma

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$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

That is: to an agent of S , *any* recipient's completed bundle looks at least one unit worse than her own.

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Strict case. If $c_i(X_i) < c_i(X_t)$, integrality gives $c_i(X_t) \geq c_i(X_i) + 1$, and monotonicity gives $c_i(X_t \cup \{e\}) \geq c_i(X_t)$. Done.

The key lemma

Lemma

Let $i \in S$, $t \in T$ and $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

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Equality case. Suppose $c_i(X_i) = c_i(X_t)$.

- If $i = t$: terminal lemma (i) gives $c_i(e \mid X_i) = 1$.
- If $i \neq t$: then $(i, t) \in E$, and $i, t \in S$, so terminal lemma (ii) gives $c_i(e \mid X_t) = 1$.

Either way $c_i(X_t \cup \{e\}) = c_i(X_i) + 1$. □

Envy-freeness of the completion

Theorem

With $\mathcal{A}$ and p as constructed, $(\mathcal{A}, p)$ is an envy-free solution.

Fix i . Since $X_j \subseteq A_j$ for every j , monotonicity and envy-freeness of X give the standing inequality

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i) \quad \text{for every } j. \quad (*)$$

We must check $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for every j . Three cases, according to where i sits:

$$\begin{array}{ccc} \underbrace{i \in T}_{\text{recipient}} & \underbrace{i \in P \setminus T}_{\text{paid, not a recipient}} & \underbrace{i \notin P}_{\text{unpaid}} \end{array}$$

(These are exhaustive: N is the disjoint union of T , $P \setminus T$ and $N \setminus P$.)

Case 1: $i \in T$ is a recipient

Then $p_i = 1$, and if $e \in R$ is the chore she absorbed, terminal lemma (i) gives

$$c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1, \quad \text{so} \quad c_i(A_i) - p_i = c_i(X_i).$$

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So we need $c_i(A_j) - p_j \geq c_i(X_i)$ for every j .

$j \notin P$ Then $j \notin T$, so $A_j = X_j$ and $p_j = 0$. Envy-freeness of X :

$$c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i). \quad \checkmark$$

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$j \in P$ Here $p_j = 1$, so we need $c_i(A_j) \geq c_i(X_i) + 1$.

- If $j \in T$: this is exactly the key lemma.
- If $j \in P \setminus T$: then $j \notin S$ by the closure lemma, and $i \in S$, so terminal lemma (iii) gives $c_i(X_i) < c_i(X_j)$; integrality then gives $c_i(A_j) = c_i(X_j) \geq c_i(X_i) + 1$. $\checkmark$

Cases 2 and 3

Case 2: $i \in P \setminus T$ Here $A_i = X_i$ and $p_i = 1$, so $c_i(A_i) - p_i = c_i(X_i) - 1$.

For every j , the standing inequality (*) and $p_j \leq 1$ give

$$c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i. \checkmark$$

(Being paid without taking on extra work is comfortable.)

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Case 3: $i \notin P$ Then $i \notin T$, so $A_i = X_i$, $p_i = 0$, and $c_i(A_i) - p_i = c_i(X_i)$.

If $j \notin P$: $A_j = X_j$, $p_j = 0$, and envy-freeness of X closes it.

If $j \in P$, we claim $c_i(A_j) \geq c_i(X_i) + 1$:

- $i \notin S$. Envy-freeness gives $c_i(X_i) \leq c_i(X_j)$. Equality would put $(i, j) \in E$; with $i \notin S$ and $j \in P$, rule (P2) would force $i \in P$ — contradiction. So the inequality is strict, and integrality with monotonicity gives the claim.
- $i \in S$. If $j \in T$ this is the key lemma. If $j \in P \setminus T$ then $j \notin S$ by the closure lemma, and terminal lemma (iii) plus integrality give the claim.

Since $p_j = 1$, we get $c_i(A_j) - p_j \geq c_i(X_i)$. $\checkmark$

□

EF1 comes for free

Corollary

$\mathcal{A}$ is EF1 — before any subsidy is paid. Precisely:

- (i) every non-recipient $i \in N \setminus T$ is *envy-free* in $\mathcal{A}$;
- (ii) every recipient t_k satisfies $c_{t_k}(A_{t_k} \setminus \{e_k\}) \leq c_{t_k}(A_j)$ for every j .

PROOF. Recall X is envy-free and $X_j \subseteq A_j$ for every j .

(i) For $i \notin T$ we have $A_i = X_i$, so for every j

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using envy-freeness of X and then monotonicity.

(ii) For $i = t_k$ we have $A_i = X_i \cup \{e_k\}$, so $A_i \setminus \{e_k\} = X_i$, and the same two steps give

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Removing the single chore she absorbed erases all her envy.

□

Putting it together

PROOF OF THE MAIN THEOREM. Pass to costs $c_i = -v_i$, dichotomous. Run the Tao et al. algorithm; let X be its terminal partial allocation, R the residue, $r = |R|$.

$r = 0$ X is already complete and envy-free, so $(X, \mathbf{0})$ is an envy-free solution with zero subsidy — and every envy-free allocation is EF1.

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$r \geq 1$ Build G , take the tail SCC S , and construct $T, \mathcal{A}, p$. Then

- $\mathcal{A}$ is complete;
- $(\mathcal{A}, p)$ is envy-free (envy-freeness theorem)
- $p \in \{0, 1\}^n, \sum_i p_i \leq n - 1$ (budget)
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- $\mathcal{A}$ is EF1 (corollary)

Running time Tao et al. is polynomial. G has n vertices and is read off the n^2 values $c_i(X_j)$ — one oracle query each. SCCs, hence a tail component, in polynomial time. Completing $\mathcal{A}$ costs $r \leq n - 1$ assignments, and P is a backward reachability search in $O(n^2)$. $\square$

Why not just mirror the goods construction?

The goods-side algorithm of Barman et al. is *incremental*: allocate items one at a time, maintaining the invariant that the partial allocation is envy-freeable with subsidies in $\{0, 1\}^n$.

The obvious chore analogue: give each new chore to an agent who currently receives *no* subsidy.

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This does not work.

There are negative dichotomous instances where a partial allocation admits an envy-free solution with subsidies in $\{0, 1\}^n$, yet after one more chore arrives *no* assignment of it preserves the unit-subsidy guarantee — even if the already-allocated bundles may first be permuted among the agents.

So the one-item extension principle fails for chores, and a different argument is needed. Here is the witness.

The instance

Three agents $N = \{1, 2, 3\}$, three chores $M = \{x, y, z\}$. Costs depend only on bundle size:

$$c_1(S) = \max \{0, |S| - 1\}, \quad c_2(S) = c_3(S) = \min \{|S|, 2\}.$$

$ S $	0	1	2	3
$c_1(S)$	0	0	1	2
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Consider the partial allocation

$$X_1 = \{x, y\}, \quad X_2 = X_3 = \emptyset,$$

with z still unallocated.

The state is legitimately reachable

$$(\emptyset, \emptyset, \emptyset) \longrightarrow (\{x\}, \emptyset, \emptyset) \longrightarrow (\{x, y\}, \emptyset, \emptyset)$$

with pointwise-minimal subsidies

$$(0, 0, 0), \quad (0, 0, 0), \quad (1, 0, 0)$$

respectively — all within $\{0, 1\}^3$. So an incremental algorithm can legitimately arrive here.

At X : agent 1 costs her bundle at 1 and either empty bundle at 0, so she needs one unit; agents 2, 3 cost their own bundles at 0 and X_1 at 2, so they have no envy.

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Claim

z cannot be inserted so that the result admits an envy-free subsidy vector in $\{0, 1\}^3$ — even allowing $\{x, y\}, \emptyset, \emptyset$ to be permuted among the agents first.

The three ways to place z

(a) z joins the two-chore bundle Bundle sizes $(3, 0, 0)$. Every agent values a three-chore bundle at 2 and an empty bundle at 0, so whoever holds it needs a subsidy at least **two** units above an agent holding nothing. Impossible in $\{0, 1\}^3$.

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(b) z joins an empty bundle, and an agent of $\{2, 3\}$ holds the pair Bundle sizes $(2, 1, 0)$. That agent values her pair at 2 and the empty bundle at 0 — again a gap of **two**. Impossible.

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(c) z joins an empty bundle, and agent 1 holds the pair The only remaining option. Let $i \in \{2, 3\}$ take the singleton and j the empty bundle. Then

$$\underbrace{1 - p_1 \leq 0 - p_i}_{\text{agent 1 vs } i} \implies p_1 \geq p_i + 1, \quad \underbrace{1 - p_i \leq -p_j}_{\text{agent } i \text{ vs } j} \implies p_i \geq p_j + 1.$$

Chaining: $p_1 \geq p_j + 2$ — impossible for $p \in \{0, 1\}^3$. □

The instance itself is fine

The obstruction is to the *method*, not to the instance.

The complete allocation $(\{x\}, \{y\}, \{z\})$ is envy-free **with no subsidy at all**.

Indeed agent 1 values every singleton at cost 0, and agents 2 and 3 value every singleton at cost 1 — nobody envies anybody.

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What went wrong. Reaching that allocation from $(\{x, y\}, \emptyset, \emptyset)$ requires *taking back* chores already placed on agent 1. An incremental construction that only ever adds items cannot do it.

This is why our proof starts from a *global* object — the terminal state of the Tao et al. algorithm — rather than building up one chore at a time.

Limitation 2: Pareto optimality

Pareto optimality is judged on the *chore allocation*, ignoring subsidies.

$\mathcal{B}$ *Pareto dominates* $\mathcal{A}$ if $c_i(\mathcal{B}_i) \leq c_i(\mathcal{A}_i)$ for every i , strictly for some. $\mathcal{A}$ is *Pareto optimal* if nothing dominates it.

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Proposition

There are negative dichotomous instances — already with two agents and identical binary *submodular* costs — in which *no* Pareto-optimal allocation admits an envy-free subsidy vector in $\{0, 1\}^n$.

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Proposition

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The instance. Two agents, three chores $\{a, b, c\}$, both with

$$c_1(S) = c_2(S) = \min \{|S|, 2\}.$$

Marginals in $\{0, 1\}$, so dichotomous. Submodular, since $k \mapsto \min \{k, 2\}$ is non-decreasing and concave. Not additive: $c_i(\{a, b, c\}) = 2 < 3 = \sum_g c_i(\{g\})$.

The clash

Up to symmetry every complete allocation has bundle sizes $(3, 0)$ or $(2, 1)$, giving cost profiles

$$(2, 0), \quad (2, 1), \quad (1, 2), \quad (0, 2).$$

The concentrated allocations are Pareto optimal None of $(2, 1)$, $(1, 2)$, $(0, 2)$ dominates $(2, 0)$ — each either raises some agent's cost or matches it without improving the other. Symmetrically for $(0, 2)$.

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And the affordable allocations are dominated A split $(2, 1)$ has profile $(2, 1)$ and is envy-free with $(1, 0)$ — but handing the third chore to the agent already holding two moves the profile to $(2, 0)$: her cost is unchanged, the other's strictly falls. So it is dominated. □

What the two limitations say

No incremental extension

The goods-side proof technique does not transfer. One must reason about a whole terminal state, not a growing prefix.

A method obstruction — the instance itself was perfectly well behaved.

No free Pareto optimality

The unit-subsidy guarantee cannot be strengthened to demand Pareto optimality of the allocation.

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Note the price of insisting on unit subsidies on that instance: the cheapest admissible allocation has total cost 3, against 2 for the cheapest Pareto-optimal one.

So efficiency and bounded subsidy genuinely pull against each other here.

Summary

Result

Negative dichotomous chores admit a complete allocation $\mathcal{A}$ and subsidies $p \in \{0, 1\}^n$ with $\sum_i p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is envy-free; $\mathcal{A}$ is EF1 before subsidies; all computable in polynomial time.

The shape of the argument.

1. Take the terminal state of the Tao et al. algorithm — envy-free, at most $n - 1$ chores left.
2. Its halting condition hands us a tail SCC S with $r < |S|$: *more agents than leftover chores*.
3. Park the leftovers on distinct agents $T \subseteq S$; each pays +1.
4. Pay T , propagating backwards along equality arcs — never into S .
5. $S \setminus T \neq \emptyset$ stays unpaid: exactly the $n - 1$ budget.

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Two boundaries. No incremental extension; no free Pareto optimality.

Directions

- **General binary marginals.** Marginals in $\{-1, 0, 1\}$ — goods and chores together, without additivity. Does the unit-subsidy guarantee survive? Open even for $n = 3$.
- **Beyond binary.** Marginals in $\{-k, \dots, 0\}$: what is the right per-agent bound?
- **Efficiency.** Our Proposition rules out Pareto optimality with unit subsidies. What is the best achievable trade-off — the least total cost among allocations admitting $p \in \{0, 1\}^n$?
- **Truthfulness.** Goko et al. obtain truthfulness for matroidal goods under free disposal. Is anything comparable possible for dichotomous chores?

Thank you

Questions welcome.

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